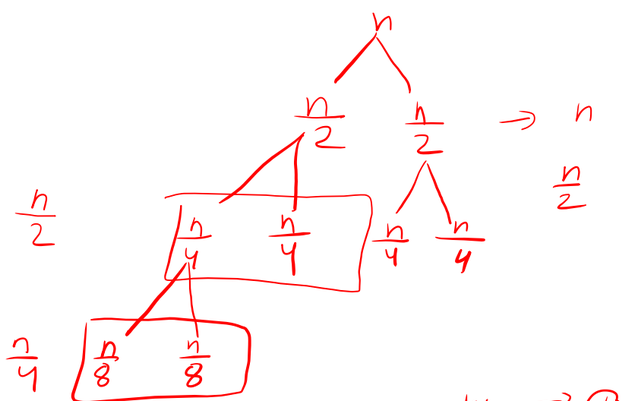
$2T\left(\frac{n}{2}\right) + n$



$\rightarrow n$

$\frac{n}{2}$

$\frac{n}{2}$

$\frac{n}{4}$

$\frac{4n}{4} = n$

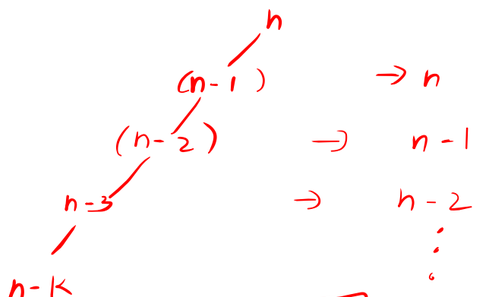
$kn \rightarrow$ ①

$\frac{n}{2^k} = 1$

$n = 2^k$ $= \log_2 n = k$ ($\log n$)

$O(\log(n) \times n)$

$T(n-1) + n$



$\rightarrow n$

$\rightarrow n-1$

$\rightarrow n-2$

$n = k$

$\sum_{i=0}^{k} i$

$1+2+3 \cdots + (n)$

$\frac{n(n+1)}{2} = \frac{n^2+n}{2}$

$O(n^2)$